Issue
5

Reports on ICON

The Tracer Transport Module Part I: A Mass Consistent Finite Volume Approach with Fractional Steps

Daniel Reinert

December 2020

DOI: 10.5676/DWD_pub/nwv/icon_005
ISSN: 2628-4898

Deutscher Wetterdienst
Wetter und Klima aus einer Hand

Max-Planck-Institut
für Meteorologie

DOI: 10.5676/DWD_pub/nwv/icon_005
ISSN: 2628-4898

The CC license “BY-NC-ND” allows others only to download the publication and share it with others as long as they credit the publication, but they can’t change it in any way or use it commercially.

Publisher

Deutscher Wetterdienst
Business Area “Research and Development”
Frankfurter Straße 135
63067 Offenbach
www.dwd.de

Editors

Sebastian Rast, MPI-M
sebastian.rast@mpimet.mpg.de
René Redler, MPI-M
rene.redler@mpimet.mpg.de
Daniel Reinert, DWD,
daniel.reinert@dwd.de
Daniel Rieger, DWD,
daniel.rieger@dwd.de
Florian Prill, DWD,
florian.prill@dwd.de

The Tracer Transport Module Part I: A Mass Consistent Finite Volume Approach with Fractional Steps

Daniel Reinert

Deutscher Wetterdienst

This document refers to release icon-2.6.0.

Abstract

The tracer transport module of ICON is presented and the consistency with the continuity equation for air mass is discussed. The transport module solves the continuity equation for tracer mass and, as such, describes the large-scale transport of gaseous, liquid or solid atmospheric constituents due to air motion. Finite volume methods are applied and combined with the fractional step approach (operator splitting) in order to derive mathematically simpler subproblems for the horizontal and vertical directions. The mathematical derivation of the discretized equation set is presented. Specific measures are taken in order to achieve consistency with continuity (CWC). These include the reuse of time-averaged mass fluxes from the discretized air mass continuity equation, as well as the re-integration of the air mass continuity equation as part of the fractional step approach. Results of an ICON forecast run are presented in order to highlight that CWC is maintained, provided that the aforementioned measures are taken.

Keywords: transport, continuity equation, finite volume, fractional steps, consistency with continuity

Correspondence to: Daniel Reinert (daniel.reinert@dwd.de)

1. Introduction

The transport module is an important building block of any numerical weather prediction (NWP) or climate model, as it predicts the large-scale redistribution of water substances, chemical constituents or aerosols in the atmosphere due to air motion. Mathematically, it solves one of the fundamental laws of physics, namely the equation of tracer mass continuity.

When developing a transport module for the atmosphere one is faced with several, partially contradicting design objectives. One major challenge is posed by the contrasting requirements of numerical accuracy on the one hand and computational efficiency (in terms of time-to-solution and memory consumption) on the other hand. As the total number of prognostic tracers can reach several dozen or even exceed 100 if atmospheric chemistry is taken into account, the transport module can easily become one of the most time consuming parts of a NWP or climate model. A

pragmatic trade-off between accuracy and efficiency is thus unavoidable. On top of that there exist many more important design objectives, such as high parallel efficiency, mass conservation, consistency between tracer mass and air mass, or consistency among different tracers. See e.g. Lauritzen et al. (2011b) for an extended list.

Part I of this report series aims to present the underlying equations in continuous form and to provide an introduction into the numerical discretization and solution procedure based on the finite volume approach. We start in Section 2 with a description of the continuous form of the tracer mass continuity equation and the connection to its nearest relative, the continuity equation for air mass. Section 3 deals with the cell-integrated form, which forms the basis for ICON's finite volume discretization. This is followed in Section 4 by a description of the basic solution procedure based on fractional steps. Details of the vertical and horizontal transport operators will be presented in Part II and III of this report series.

2. The Tracer Mass Continuity Equation

In mathematical terms, the tracer mass continuity equation is a first-order linear hyperbolic partial differential equation with source terms. For an arbitrary species χ it reads in flux form

$$\frac{\partial \rho q_\chi}{\partial t} + \nabla \cdot (\rho q_\chi \, \boldsymbol{v}) = Q_{\rho q_\chi} \,, \tag{1}$$

where ρ denotes the air density, $\boldsymbol{v}$ is the 3D velocity vector, t is the time, $\nabla\cdot$ represents a three-dimensional flux divergence, and q_χ denotes the tracer mass fraction. In ICON, the prognostic (moist) air density ρ is the sum of the dry air density ρ_d, the density of water vapour, and all condensates. Hence, q_χ is defined wrt moist air density as follows

$$q_\chi = \frac{\rho_\chi}{\rho} = \frac{\rho_\chi}{\sum_k \rho_k} \,,$$

where ρ_χ is the partial density of tracer species χ. Summation over all atmospheric constituents leads to the constraint $\sum_k q_k = 1$. $Q_{\rho q_\chi}$ in (1) represents sources and sinks of ρq_χ, due to, e.g., chemical or microphysical processes. $Q_{\rho q_\chi}$ is shorthand for

$$Q_{\rho q_\chi} = \left.\frac{\partial \rho q_\chi}{\partial t}\right|_{phy} = \rho \left.\frac{\partial q_\chi}{\partial t}\right|_{phy} + q_\chi \left.\frac{\partial \rho}{\partial t}\right|_{phy} \,,$$

so ρq_χ is subject to changes if any physical process causes changes in ρ or q_χ, or both.

By choosing $q_\chi = 1$ in (1) we recover the continuity equation for moist air

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \, \boldsymbol{v}) = Q_\rho \,, \qquad \text{with} \quad Q_\rho = \left.\frac{\partial \rho}{\partial t}\right|_{phy} = \sum_k Q_{\rho q_k} \,. \tag{2}$$

From (1) and (2) it becomes clear that, for any type of flow, an initially uniform q field (e.g. $q(x, y, z, t_0) = 1$) remains uniform in the absence of q-sources and sinks. This property is oftentimes referred to as *tracer and air mass consistency* or *consistency with continuity* (CWC, Gross et al., 2002). Care must be taken that this property is also valid for the discretized continuity equation. Lin and Rood (1996) give the following convenient definition of CWC for numerical discretizations:

A discretization of the advection equation is consistent with continuity if, given a spatially uniform scalar field as an initial datum, and a general flow field, the discretized scalar advection equation reduces to the discretized continuity equation.

2.1. Mass Continuity Equations in ICON

The total number of prognostic continuity equations in ICON depends on the model setup. In DWD's global operational applications, ICON solves continuity equations of type (1) for 5 different water species: water vapour q_v, cloud water q_c, rain water q_r, cloud ice q_i and snow q_s. In high resolution limited area applications (ICON-D2, see DWD Database Reference (2020)) an additional continuity equation is solved for graupel q_g. Until today, the cutting edge simulation at DWD in terms of prognostic tracers is the global ICON-ART simulation of mineral dust, which in total solves 11 continuity equations (5 water species and 3 mineral dust tracers with one prognostic equation for mass concentration and number concentration, each).

The term $Q_{\rho q_\chi}$ in (1) denotes the sum of sources and sinks due to various physical processes such as

- saturation adjustment $Q^{sat}_{\rho q_\chi}$
- grid scale microphysics $Q^{gsm}_{\rho q_\chi}$
- convection $Q^{con}_{\rho q_\chi}$
- turbulent transport $Q^{turb}_{\rho q_\chi}$
- surface fluxes $Q^{sfc}_{\rho q_\chi}$
- incremental analysis update (IAU) $Q^{iau}_{\rho q_\chi}$

Strictly speaking, $Q^{iau}_{\rho q_\chi}$ is not a physical tendency. It is merely a forcing term through which the model is pulled towards the analyzed state in operational applications.

In ICON, the specific treatment of physics tendencies wrt computation and time integration depends on the type of the underlying physical process. We distinguish between so called *fast* and *slow* physical processes. Fast physics are treated with the sequential-update split method (aka time split/fractional steps), whereas slow physics are treated with the parallel split method. These methods differ in both the physical state that enters the computation of a tendency as well as the way the tendency is applied to update the physical state. See e.g. Donahue and Caldwell (2018) for a convenient description of these splitting methods.

Note that fast physics tendencies will be ignored in the following. Whenever the terms $Q_{\rho q_\chi}$ or Q_ρ appear, it is implicitly assumed that they refer to slow physics tendencies such as convection. For additional details regarding slow and fast physics, the reader is referred to the documentation of the NWP physics package in the ICON model tutorial (Prill et al., 2020).

3. Finite Volume Discretization

The numerical solution to the tracer mass continuity equation (1) is based on so called space-time finite volume methods. By space-time methods we refer to methods, where the temporal and

spacial discretizations are combined rather than separated. Space-time methods are also known as *cell-integrated semi-Lagrangian* schemes. As will become clear, such schemes are neither purely semi-Lagrangian nor Eulerian in a strict sense. They are Eulerian in the sense that the flux of mass through the stationary walls of grid cells is considered. They are, however, semi-Lagrangian in the sense that trajectory calculations are performed for flux computation. The schemes that will be presented in this work could be termed *flux form semi-Lagrangian*. They build upon work by Lauritzen et al. (2010, 2011a), Harris and Lauritzen (2010), Skamarock and Menchaca (2010), Miura (2007) for the horizontal and Colella and Woodward (1984), Zerroukat et al. (2006) for the vertical.

It is to be noted that different numerical methods are used for tracer mass (1) and air mass (2). Moreover, different timesteps are used. The maximum allowable time step for the integration of the air mass continuity equation is determined by the speed of the fastest wave in the system - the speed of sound. The maximum allowable time step for the tracer mass continuity equation, however, is determined by the wind speed. As the wind speed is typically small compared to the speed of sound (at least in large parts of the atmosphere), on default a five times larger time step is used for tracer mass than for air mass (see also Section 3.3). Sometimes this is referred to as *air mass sub-stepping*.

3.1. Grid and Notation

The finite volume discretization in horizontal directions builds on the standard icosahedral-triangular grid with C-grid staggering, which is also used for the momentum and thermodynamic equations in the dynamical core (Zängl et al., 2015). The velocity component normal to the cell edges $v_{n\,e,k}$ is defined at edge midpoints and the tracer mass fraction $q_{i,k}$ is defined at cell circumcenters. The indices i and k denote the horizontal cell and vertical (full) level, respectively, while e denotes the cell's edge.

In vertical direction, a height-based, stationary grid with Lorenz-type staggering is used. $q_{i,k}$ is defined at full levels, while vertical velocity w is staggered wrt q and stored at half levels. Vertical levels are numbered in top-down order. Thus $k-1/2$ and $k+1/2$ denote the upper and lower vertical half-level, respectively. Taking a 3D view, the discretization relies on prismatic cells (also called *control volumes*), with w defined at the top and bottom faces, v_n at the lateral faces and q in the center (see Fig. 1 in Wan et al. (2013)).

Since we are dealing with a finite volume discretization it is worth noting that in the remainder of this work all discrete scalar variables $\psi_{i,k}$, whose storage location is at the circumcenter of an arbitrary control volume $V_{i,k}$, are interpreted as cell averages rather than point values, i.e.

$$\overline{\psi}_{i,k} = \frac{1}{\Delta V_{i,k}} \iiint\limits_{V_{i,k}} \psi(x,y,z,t)\,\mathrm{d}V\,,$$

with $\Delta V_{i,k}$ denoting the volume of $V_{i,k}$ and the overbar indicating the cell average. $\psi(x,y,z,t)$ denotes the sub-grid distribution, which is generally unknown.

3.2. Cell-Integrated Version of the Continuity Equation

Finite volume methods are derived on the cell integrated form of the underlying PDE. A cell-integrated solution to the continuity equation (1) can formally be derived, by integrating in space over the 3D prismatic control volume $V_{i,k}$ and in time over the time interval $[t^n, t^{n+1}]$. Assuming a rectangular basis, and that the top and bottom faces of $V_{i,k}$ are of identical size, one arrives at

$$\begin{aligned}\int_{t^n}^{t^{n+1}} \iint_{A_i} \int_{z_{k+\frac{1}{2}}}^{z_{k-\frac{1}{2}}} \frac{\partial(\rho\, q)}{\partial t}\, \mathrm{d}t\, \mathrm{d}A\, \mathrm{d}z = &- \int_{t^n}^{t^{n+1}} \iint_{A_i} \int_{z_{k+\frac{1}{2}}}^{z_{k-\frac{1}{2}}} \nabla_h \cdot (\rho\, q \boldsymbol{v}_h)\, \mathrm{d}t\, \mathrm{d}A\, \mathrm{d}z \\ &- \int_{t^n}^{t^{n+1}} \iint_{A_i} \int_{z_{k+\frac{1}{2}}}^{z_{k-\frac{1}{2}}} \frac{\partial}{\partial z} (\rho\, q w)\, \mathrm{d}t\, \mathrm{d}A\, \mathrm{d}z \\ &+ \int_{t^n}^{t^{n+1}} \iint_{A_i} \int_{z_{k+\frac{1}{2}}}^{z_{k-\frac{1}{2}}} Q_{\rho q}\, \mathrm{d}t\, \mathrm{d}A\, \mathrm{d}z\,.\end{aligned} \tag{3}$$

Here, A_i denotes the 2D domain which results when projecting $V_{i,k}$ onto the horizontal plane, $\Delta z_k = z_{k-1/2} - z_{k+1/2}$ is the cell thickness and the subscript h indicates horizontal 2D operators and vectors. The subscript χ has been omitted for brevity. Please note that (3) is only strictly valid if the shallow atmosphere approximation is used. Otherwise the domain A_i is a function of z.

Commuting temporal differentiation and spatial integration on the LHS as well as horizontal/vertical differentiation with vertical/horizontal integration on the RHS leads to

$$\begin{aligned}(\overline{\rho q}\, \Delta A\, \Delta z)_{i,k}^{n+1} - (\overline{\rho q}\, \Delta A\, \Delta z)_{i,k}^{n} = &- \int_{t^n}^{t^{n+1}} \iint_{A_i} \nabla_h \cdot (\overline{\rho q \boldsymbol{v}_h}^z \Delta z)_k\, \mathrm{d}A\, \mathrm{d}t \\ &- \int_{t^n}^{t^{n+1}} \int_{z_{k+\frac{1}{2}}}^{z_{k-\frac{1}{2}}} \frac{\partial}{\partial z} \left(\overline{\rho q w}^h \Delta A\right)_i\, \mathrm{d}z\, \mathrm{d}t \\ &+ \int_{t^n}^{t^{n+1}} \overline{Q}_{\rho q;i,k} \Delta A_i \Delta z_k\, \mathrm{d}t\,.\end{aligned}$$

The superscripts $\overline{}^z$, $\overline{}^h$ denote vertical and horizontal averages, respectively, for the grid cell at hand, ΔA_i is the area of domain A_i (also called the *cell area*), and $\overline{Q}_{\rho q;i,k}$ denotes the cell averaged source/sink. Since the shape of our grid cell is independent of time, the validity of this commutation is readily verifiable for the LHS (Leibniz integral rule). The interchange of vertical differentiation and horizontal integration for the second term on the RHS implies the assumption that the integration bounds are independent of z[1]. An equivalent assumption applies to the first term on the RHS.

[1] Note that this is only strictly valid if the shallow atmosphere approximation is used.

Applying the Gauss-Theorem and the fundamental law of calculus on the RHS leads to

$$
\begin{aligned}
(\overline{\rho q}\,\Delta A\,\Delta z)_{i,k}^{n+1} - (\overline{\rho q}\,\Delta A\,\Delta z)_{i,k}^{n} = &- \int_{t^n}^{t^{n+1}} \oint_{\partial A_i} (\overline{\rho q \boldsymbol{v}_h}^{z} \Delta z)_k \cdot \mathbf{n}\,\mathrm{d}l\,\mathrm{d}t \\
&- \int_{t^n}^{t^{n+1}} \left[\left(\overline{\rho q w}^{h} \Delta A\right)_{i,k-\frac{1}{2}} - \left(\overline{\rho q w}^{h} \Delta A\right)_{i,k+\frac{1}{2}} \right] \mathrm{d}t \\
&+ \int_{t^n}^{t^{n+1}} \overline{Q}_{\rho q;i,k} \Delta A_i \Delta z_k \,\mathrm{d}t\,,
\end{aligned}
\tag{4}
$$

where $\mathbf{n}$ is the outward pointing normal vector along the boundary of A_i. Noting that the integral along the boundary ∂A_i can be split into a sum of integrals along the N_e triangle edges, equation (4) can be modified as follows:

$$
\begin{aligned}
(\overline{\rho q}\,\Delta A\,\Delta z)_{i,k}^{n+1} - (\overline{\rho q}\,\Delta A\,\Delta z)_{i,k}^{n} = &- \sum_{e=1}^{N_e} \int_{t^n}^{t^{n+1}} \int_{l_e} (\overline{\rho q \boldsymbol{v}_h}^{z} \Delta z \cdot \mathbf{n})_{e,k}\,\mathrm{d}l\,\mathrm{d}t \\
&- \int_{t^n}^{t^{n+1}} \left[\left(\overline{\rho q w}^{h} \Delta A\right)_{i,k-\frac{1}{2}} - \left(\overline{\rho q w}^{h} \Delta A\right)_{i,k+\frac{1}{2}} \right] \mathrm{d}t \\
&+ \int_{t^n}^{t^{n+1}} \overline{Q}_{\rho q;i,k} \Delta A_i \Delta z_k \,\mathrm{d}t
\end{aligned}
$$

It is to be noted that each term on the RHS (except the last one) represents the mass transported through a single face of our prismatic cell during the time step $\Delta t = t^{n+1} - t^n$. The task of any finite volume scheme is to find an adequate approximation to this time integrated mass.

Before we move on, we simplify our prognostic equation a bit further by noting that ΔA_i is independent of time as well as independent of the vertical level k[2]. Hence, division by ΔA_i gives:

$$
\begin{aligned}
(\overline{\rho q}\,\Delta z)_{i,k}^{n+1} - (\overline{\rho q}\,\Delta z)_{i,k}^{n} = &- \frac{1}{\Delta A_i} \sum_{e=1}^{N_e} \int_{t^n}^{t^{n+1}} \int_{l_e} (\overline{\rho q \boldsymbol{v}_h}^{z} \Delta z \cdot \mathbf{n})_{e,k}\,\mathrm{d}l\,\mathrm{d}t \\
&- \int_{t^n}^{t^{n+1}} \left[\left(\overline{\rho q w}^{h}\right)_{i,k-\frac{1}{2}} - \left(\overline{\rho q w}^{h}\right)_{i,k+\frac{1}{2}} \right] \mathrm{d}t \\
&+ \Delta t \Delta z_k \widetilde{Q}_{\rho q;i,k}
\end{aligned}
\tag{5}
$$

In addition we have introduced the time averaged source/sink term

$$
\widetilde{Q}_{\rho q;i,k} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} \overline{Q}_{\rho q;i,k}\,\mathrm{d}t\,.
$$

[2]This assumption is only valid if the shallow atmosphere approximation is used.

Now, each term on the RHS of (5) (except the last one) represents the time integrated mass flux through one face. In general, these terms cannot be computed analytically and, hence, must be approximated numerically.

3.3. Time-averaged Mass Fluxes

A necessary condition for CWC is that the same mass flux is 'seen' by the air mass continuity equation and the tracer mass continuity equation. The most straightforward way to achieve this is to use the same numerical methods for (1) and (2). Ideally, however, one would like to keep the possibility to choose different numerical methods for ρ and ρq without sacrificing CWC. Specific numerical properties might be more important for tracers than for air mass (and vice versa). For example, monotonicity is an important property for tracers, while it is probably less crucial for air mass. Thus, one might be tempted to choose a monotonic numerical scheme for tracers and a non-monotonic one for air mass.

An alternative to using identical numerical methods is to enforce identical mass fluxes. The basic idea is as follows: The mass continuity equation (2) is solved prior to (1) with a numerical method of choice, and the mass fluxes are stored in memory. In a second step, these mass fluxes are reused for solving the prognostic tracer mass equation (1). With this method CWC can be ensured, even if different numerical methods are applied. It is even possible to use larger integration time steps for (1) than for (2) by storing time-averaged rather than instantaneous mass fluxes. ICON makes use of this concept, as detailed in the following:

Starting from the cell-integrated continuity equation (5) we approximate the instantaneous, spatially averaged mass fluxes as follows:

$$\overline{\rho q \boldsymbol{v}_h}^z \approx \overline{\rho \boldsymbol{v}_h}^z \overline{q}^z$$
$$\overline{\rho q w}^h \approx \overline{\rho w}^h \overline{q}^h$$

Further we assume that the mass fluxes $\overline{\rho \boldsymbol{v}_h}^z$ and $\overline{\rho w}^h$ are constant in time during Δt and that $\overline{\rho \boldsymbol{v}_h}^z$ is constant along the edge e. With these assumptions both mass fluxes can be moved outside of the time integral and the horizontal mass flux can furthermore be moved outside of the line integral along the edge. Finally this leads to

$$\begin{aligned}(\overline{\rho q}\,\Delta z)^{n+1}_{i,k} - (\overline{\rho q}\,\Delta z)^{n}_{i,k} = &- \frac{1}{\Delta A_i}\sum_{e=1}^{N_e}(\overline{\rho \boldsymbol{v}_h}^z \Delta z \cdot \mathbf{n})_{e,k}\int_{l_e}\int_{t^n}^{t^{n+1}} \overline{q}^z_{e,k}\,\mathrm{d}t\,\mathrm{d}l \\ &- \left[\overline{\rho w}^h_{i,k-1/2}\int_{t^n}^{t^{n+1}} \overline{q}^h_{i,k-1/2}\,\mathrm{d}t - \overline{\rho w}^h_{i,k+1/2}\int_{t^n}^{t^{n+1}} \overline{q}^h_{i,k+1/2}\,\mathrm{d}t\right] \\ &+ \Delta t \Delta z_{i,k}\widetilde{Q}_{\rho q;i,k}\end{aligned} \tag{6}$$

The basic idea now is to replace $\overline{\rho \boldsymbol{v}_h}^z$ and $\overline{\rho w}^h$ by mass fluxes which are already known from the solution of the air mass continuity equation (2). If the air mass equation is sub-cycled with respect to the tracer mass continuity equations, we define time averaged mass fluxes as follows:

Let $\Delta\tau$ denote the time step which is used for integrating the air mass continuity equation (2), with

$$\Delta\tau = \frac{1}{N_{sub}}\Delta t\,, \qquad N_{sub} \geq 1\,,$$

and a default value of $N_{sub} = 5$. Then we define the time averaged mass fluxes

$$\widetilde{\boldsymbol{F}}_{h;e,k} = \frac{1}{\Delta t\, l_e} \int_{t^n}^{t^{n+1}} \int_{l_e} \overline{\rho \boldsymbol{v}_h}^z_{e,k}\, \mathrm{d}l\, \mathrm{d}t \approx \frac{1}{\Delta t} \sum_{n=0}^{N_{sub}-1} \boldsymbol{F}^{t+n\Delta\tau}_{h;e,k}\Delta\tau$$

$$\widetilde{F}_{z;i,k-1/2} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} \overline{\rho w}^h_{i,k-1/2}\, \mathrm{d}t \approx \frac{1}{\Delta t} \sum_{n=0}^{N_{sub}-1} F^{t+n\Delta\tau}_{z;i,k-1/2}\Delta\tau\,,$$

and insert them into (6). $\boldsymbol{F}_h$ and F_z denote the known horizontal and vertical mass fluxes from the solution to the air mass continuity equation, and l_e is the length of the eth cell edge. Only the edge-normal component of $\widetilde{\boldsymbol{F}}_{h;e,k}$ is prognosed by ICON and enters the transport equation (6). Nevertheless, at this stage we keep the full 2D vector $\widetilde{\boldsymbol{F}}_{h;e,k}$, as this will be favourable for the formal derivation of the horizontal flux operator. Details will be presented in Part III of this report series. It is to be noted that in ICON N_{sub} (and Δt) can be specified by the user. Hence, the time step for air mass $\Delta\tau(\Delta t, N_{sub})$ (also called *dynamics time step*) is a derived variable.

3.4. Schematic Form

As a preparatory step for the upcoming sections, we write (6) in a more abstract form. We define the horizontal and vertical linear operators

$$\mathcal{H}(\overline{q}^z(x,y,t)) = -\frac{1}{\Delta t \Delta A_i \Delta z_{i,k}} \sum_{e=1}^{N_e} \left(\widetilde{\boldsymbol{F}}_h \cdot \mathbf{n}\,\Delta z\right)_{e,k} \int_{l_e} \int_{t^n}^{t^{n+1}} \overline{q}^z_{e,k}\, \mathrm{d}t\, \mathrm{d}l \tag{7}$$

$$\mathcal{V}(\overline{q}^h(z,t)) = -\frac{1}{\Delta t \Delta z_{i,k}} \left[\widetilde{F}_{z;i,k-1/2} \int_{t^n}^{t^{n+1}} \overline{q}^h_{i,k-1/2}\, \mathrm{d}t - \widetilde{F}_{z;i,k+1/2} \int_{t^n}^{t^{n+1}} \overline{q}^h_{i,k+1/2}\, \mathrm{d}t\right], \tag{8}$$

such that equation (6) can schematically be written as follows:

$$\overline{\rho q}^{n+1}_{i,k} = \overline{\rho q}^n_{i,k} + \Delta t \left[\mathcal{H}(\overline{q}^z) + \mathcal{V}(\overline{q}^h) + \widetilde{Q}_{\rho q}\right] \tag{9}$$

Inspecting $\mathcal{H}$ and $\mathcal{V}$, we note that the tracer transport problem has been reduced to computing the time integral and line integral of q at the faces of the prismatic control volume. Though not obvious from (7)–(9), we anticipate at this stage that in ICON the operators $\mathcal{H}$ and $\mathcal{V}$ only depend on q at time t^n and no other time. The reason will be discussed in Part II and III of this report series. As a next step, the solution procedure for equation (9) is presented.

4. Fractional Steps

In order to reduce the computational burden of solving the three-dimensional equation (9), a fractional step approach is taken in ICON (see e.g. LeVeque, 2002). Equation (9) is split into three subproblems, which can be solved independently and which successively involve the horizontal transport operator $\mathcal{H}$, the vertical transport operator $\mathcal{V}$, and the source term $\widetilde{Q}$

$$\begin{aligned}
\overline{\rho q}^{*}_{i,k} &= \overline{\rho q}^{\alpha_1}_{i,k} + \Delta t\, \mathcal{V}(\overline{q}^{\beta_1}) \\
\overline{\rho q}^{**}_{i,k} &= \overline{\rho q}^{\alpha_2}_{i,k} + \Delta t\, \mathcal{H}(\overline{q}^{\beta_2}) \\
\overline{\rho q}^{***}_{i,k} &= \overline{\rho q}^{\alpha_3}_{i,k} + \Delta t\, \widetilde{Q}^{\beta_3}_{\rho q}\ ,
\end{aligned}$$

with α, β denoting yet unspecified time levels. The superscripts h and z have been omitted for reasons of readability. If the 3D advection operator is split along coordinate directions, as it is done here, this approach is sometimes also referred to as *directional splitting*. Of course, replacing equation (9) by some combination of subproblems will inevitably result in a residual error[3]. This error is known as the *splitting error*. The following approximation to (9) is used:

$$\overline{\rho q}^{*}_{i,k} = \overline{\rho q}^{n}_{i,k} + \Delta t\, \mathcal{V}\left(\overline{q}^{n}\right) \tag{10}$$

$$\overline{\rho q}^{**}_{i,k} = \overline{\rho q}^{*}_{i,k} + \Delta t\, \mathcal{H}\left(\overline{q}^{*}\right) \tag{11}$$

$$\overline{\rho q}^{n+1}_{i,k} = \overline{\rho q}^{**}_{i,k} + \Delta t\, \widetilde{Q}^{n}_{\rho q} \tag{12}$$

This approximation is $\mathcal{O}(\Delta t)$ as $\Delta t \longrightarrow 0$, irrespective of the accuracy of the operators $\mathcal{V}$, $\mathcal{H}$ and the approximation of the time average $\widetilde{Q}_{\rho q}$. However, supposing that the individual operators $\mathcal{V}$, $\mathcal{H}$, as well as $\widetilde{Q}_{\rho q}$ are $\mathcal{O}(\Delta t^2)$, second-order accuracy can be obtained for the split scheme by using *Strang splitting* (Strang, 1968). As outlined in LeVeque (2002), Durran (1999), Strang splitting does not necessarily increase the computational burden. In essence, all that is required is to revert the order of the individual operators on alternate time steps. It is to be noted that in ICON only the advection operators are reversed. The source term $\widetilde{Q}_{\rho q}$ is always applied last during the third substep. Hence, the split scheme in ICON is not fully second-order accurate in time.

For the advection operators full Strang-splitting of the form $[\Delta t/2\, \mathcal{V}][\Delta t\, \mathcal{H}][\Delta t/2\, \mathcal{V}]$ has also been tested during the implementation phase. Except for being more expensive (the vertical operator is called twice per timestep) no significant impact on the model results has been noted.

4.1. Ensuring Consistency with Continuity

Even though we make direct use of the mass fluxes from the dynamical core (see Section 3.3), CWC can easily be destroyed by the fractional step approach. A key aspect of the splitting (10)–(12) is the computation of the intermediate tracer concentration $\overline{q}^{*}$, which enters the horizontal transport operator $\mathcal{H}$. Tempting candidates for the conversion $\overline{\rho q}^{*}_{i,k} \longrightarrow \overline{q}^{*}$ might be $\overline{\rho}^{n}$ or $\overline{\rho}^{n+1}$ as they are readily available from ICON's dynamical core. Any such attempt, however, will destroy CWC. An initially uniform tracer field will not be preserved in a deformational flow, even if the flow is non-divergent.

[3] except for the special case that the solution operators to the subproblems commute.

In order to retain CWC it is necessary to keep track of the changes in partial density ρq that are solely a result of mass convergence/divergence in the directions of splitting. Therefore we follow the method of Easter (1993), wherein the air mass continuity equation (2) is simultaneously re-integrated in the same split manner as the continuity equation for tracer mass.

ICON's solution procedure to (9) is shown in Figure 1, with

$$\mathcal{H}(1) = -\frac{1}{\Delta A_i \Delta z_{i,k}} \sum_{e=1}^{N_e} \widetilde{\boldsymbol{F}}_{h;e,k} \cdot \boldsymbol{n}_{e,k} \Delta z_{e,k}\, l_e$$

$$\mathcal{V}(1) = -\frac{1}{\Delta z_{i,k}} \left(\widetilde{F}_{z;i\,k-1/2} - \widetilde{F}_{z;i\,k+1/2} \right)$$

denoting the horizontal and vertical mass flux divergences, which rely on the precomputed, time averaged fluxes of air mass $\widetilde{\boldsymbol{F}}_h$, $\widetilde{F}_z$.

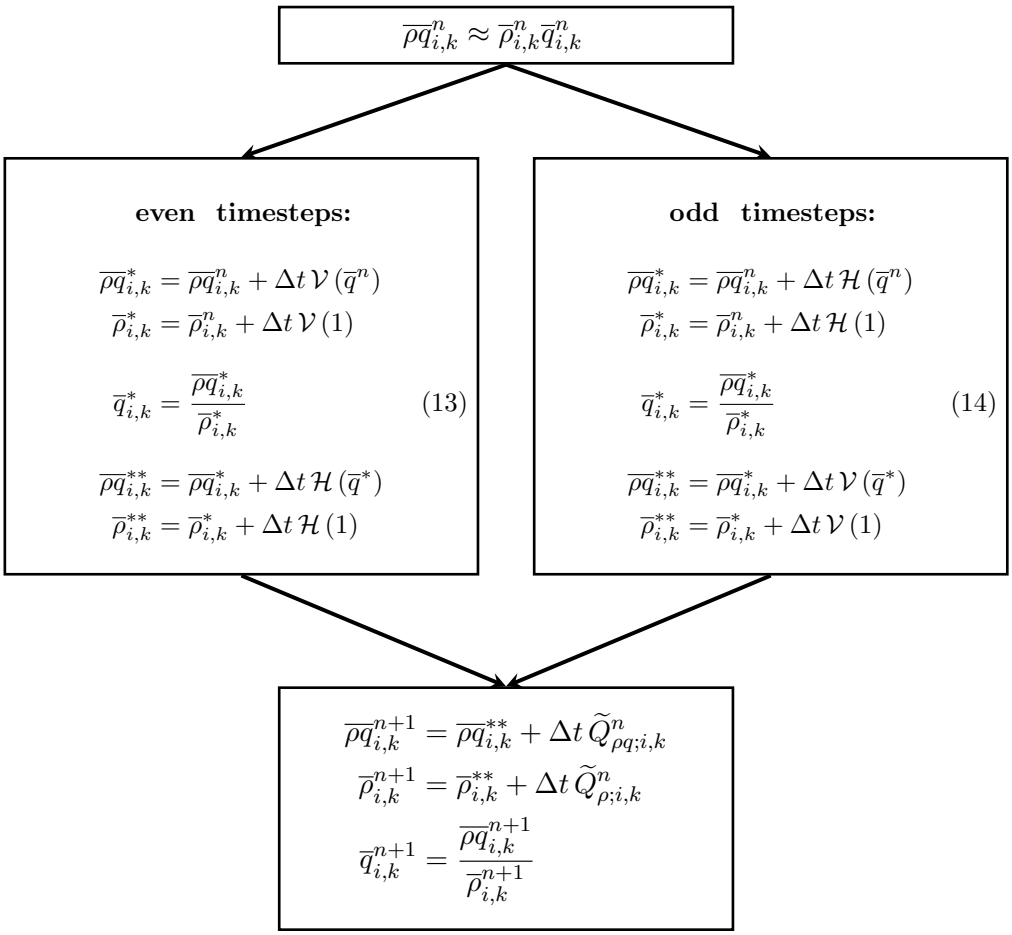

Figure 1: Fractional step method applied to the tracer mass continuity equation (9). Following Easter (1993), the air mass continuity equation is re-integrated in order to maintain CWC. The order of $\mathcal{H}$, $\mathcal{V}$ is reversed on alternate time steps Δt in order to reduce the splitting error.

The key point here is that the intermediate density $\overline{\rho}^*$ rather than $\overline{\rho}^{n+1}$ or $\overline{\rho}^n$ is used to recover

the mass fraction $\overline{q}^*$ in (13) and (14). The re-integration of (2) is rather straightforward, as it relies on the precomputed mass fluxes $\widetilde{\boldsymbol{F}}_h$, $\widetilde{F}_z$.

4.2. Implemented Form

The implemented fractional step approach differs in some important details from the standard method depicted in Figure 1. To some extent this is due to historical reasons. Initially it was assumed that the air mass continuity equation (2) in ICON always has zero RHS. Hence, grid-scale advection was assumed to be the only acting process. Based on this assumption the standard method had been implemented and slightly reformulated in order to optimize computational performance. Later on, the assumption of a zero RHS had to be relaxed (i.e. for a valid implementation of the IAU method) which then required some a posteriori corrections.

The implemented fractional step approach is given in Figure 2, with deviations from the standard method highlighted in red.

The main differences are as follows:

- The integration of the source term $\widetilde{Q}^n_{\rho q}$ is split into two parts. Right at the beginning of the integration step the term $q^n\widetilde{Q}^n_\rho = q^n\partial\rho/\partial t$ is applied, which accounts for changes in ρq due to changes in ρ (16). In parallel, the first re-integration step for ρ is performed, accounting for any non-zero RHS (15). The second term $\widetilde{Q}^n_{\rho q} - q^n\widetilde{Q}^n_\rho = \rho^n\partial q/\partial t$ which takes into account changes in ρq due to changes in q is applied at the very end of the integration (22). It is to be noted that the subtraction of $q^n\widetilde{Q}^n_\rho$ is essential in order to avoid double counting of the air mass tendency.
- Equation (19) is mathematically equivalent to
$$\overline{\rho}^{**}_{i,k} = \overline{\rho}^{*}_{i,k} + \Delta t\,\mathcal{H}\,(1)\ ,$$
but slightly more efficient, as it involves the less expensive vertical operator $\mathcal{V}(1)$ rather than $\mathcal{H}(1)$.
- Finally, the re-integration step $\overline{\rho}^{**} \longrightarrow \overline{\rho}^{n+1}$ is omitted, as $\overline{\rho}^{n+1}$ is already available from the dynamical core (see equations (18), (21)).

If we combine the fractional steps, the approximation to the tracer mass continuity equation (9) reads:

$$\overline{q}^* = \frac{\overline{\rho}^n_{i,k}\overline{q}^n_{i,k} + \Delta t\,\mathcal{V}\,(\overline{q}^n) + \Delta t\,\overline{q}^n_{i,k}\widetilde{Q}^n_{\rho;i,k}}{\overline{\rho}^n_{i,k} + \Delta t\,\mathcal{V}\,(1) + \Delta t\,\widetilde{Q}^n_{\rho;i,k}} \tag{23}$$

even:
$$\overline{\rho q}^{n+1}_{i,k} = \overline{\rho}^n_{i,k}\overline{q}^n_{i,k} + \Delta t\,\mathcal{V}\,(\overline{q}^n) + \Delta t\,\mathcal{H}\,(\overline{q}^*) + \Delta t\,\widetilde{Q}^n_{\rho q;i,k} \tag{24}$$

$$\overline{q}^* = \frac{\overline{\rho}^n_{i,k}\overline{q}^n_{i,k} + \Delta t\,\mathcal{H}\,(\overline{q}^n) + \Delta t\,\overline{q}^n_{i,k}\widetilde{Q}^n_{\rho;i,k}}{\overline{\rho}^n_{i,k} + \Delta t\,\mathcal{H}\,(1) + \Delta t\,\widetilde{Q}^n_{\rho;i,k}} \tag{25}$$

odd:
$$\overline{\rho q}^{n+1}_{i,k} = \overline{\rho}^n_{i,k}\overline{q}^n_{i,k} + \Delta t\,\mathcal{V}\,(\overline{q}^*) + \Delta t\,\mathcal{H}\,(\overline{q}^n) + \Delta t\,\widetilde{Q}^n_{\rho q;i,k} \tag{26}$$

with the re-integrated air mass continuity equation

$$\overline{\rho}^{n+1}_{i,k} = \overline{\rho}^n_{i,k} + \Delta t\,\mathcal{V}\,(1) + \Delta t\,\mathcal{H}\,(1) + \Delta t\,\widetilde{Q}^n_{\rho;i,k} \tag{27}$$

$$\overline{\rho}^{*}_{i,k} = \overline{\rho}^{n}_{i,k} + \Delta t\, \widetilde{Q}^{n}_{\rho;i,k} \tag{15}$$

$$\begin{aligned}\overline{\rho q}^{n}_{i,k} &\approx \overline{\rho}^{*}_{i,k}\overline{q}^{n}_{i,k} \\ &= \overline{\rho}^{n}_{i,k}\overline{q}^{n}_{i,k} + \Delta t\, \overline{q}^{n}_{i,k}\widetilde{Q}^{n}_{\rho;i,k}\end{aligned} \tag{16}$$

even timesteps:

$$\begin{aligned}\overline{\rho q}^{*}_{i,k} &= \overline{\rho q}^{n}_{i,k} + \Delta t\, \mathcal{V}\left(\overline{q}^{n}\right) \\ \overline{\rho}^{**}_{i,k} &= \overline{\rho}^{*}_{i,k} + \Delta t\, \mathcal{V}\left(1\right)\end{aligned}$$

$$\overline{q}^{*}_{i,k} = \frac{\overline{\rho q}^{*}_{i,k}}{\overline{\rho}^{**}_{i,k}} \tag{17}$$

$$\begin{aligned}\overline{\rho q}^{**}_{i,k} &= \overline{\rho q}^{*}_{i,k} + \Delta t\, \mathcal{H}\left(\overline{q}^{*}\right) \\ \overline{\rho}^{n+1}_{i,k} &= \cancel{\overline{\rho}^{**}_{i,k} + \Delta t\, \mathcal{H}\left(1\right)}\end{aligned} \tag{18}$$

odd timesteps:

$$\begin{aligned}\overline{\rho q}^{*}_{i,k} &= \overline{\rho q}^{n}_{i,k} + \Delta t\, \mathcal{H}\left(\overline{q}^{n}\right) \\ \overline{\rho}^{**}_{i,k} &= \overline{\rho}^{n+1}_{i,k} - \Delta t\, \mathcal{V}\left(1\right)\end{aligned} \tag{19}$$

$$\overline{q}^{*}_{i,k} = \frac{\overline{\rho q}^{*}_{i,k}}{\overline{\rho}^{**}_{i,k}} \tag{20}$$

$$\begin{aligned}\overline{\rho q}^{**}_{i,k} &= \overline{\rho q}^{*}_{i,k} + \Delta t\, \mathcal{V}\left(\overline{q}^{*}\right) \\ \overline{\rho}^{n+1}_{i,k} &= \cancel{\overline{\rho}^{**}_{i,k} + \Delta t\, \mathcal{V}\left(1\right)}\end{aligned} \tag{21}$$

$$\begin{aligned}\overline{\rho q}^{n+1}_{i,k} &= \overline{\rho q}^{**}_{i,k} + \Delta t\, \widetilde{Q}^{n}_{\rho q;i,k} - \Delta t\, \overline{q}^{n}_{i,k}\widetilde{Q}^{n}_{\rho;i,k} \\ \overline{q}^{n+1}_{i,k} &= \frac{\overline{\rho q}^{n+1}_{i,k}}{\overline{\rho}^{n+1}_{i,k}}\end{aligned} \tag{22}$$

Figure 2: Implemented form of the fractional step method. Differences to the standard method are highlighted in red. In essence it differs from the standard method by the fact that part of the physics tendency $\widetilde{Q}_{\rho q}$ which accounts for changes in ρq due to changes in ρ is already applied at the very beginning of the integration step.

Given that the operators $\mathcal{V}$, $\mathcal{H}$ maintain CWC, the fractional step approach (23)–(27) maintains CWC for odd and even time steps. This can be verified by replacing $\overline{q}^{n}$ in (23)-(26) with a constant value.

4.3. Proof of Concept

A 12 h forecast has been performed with the full-fledged ICON model in NWP-mode, in order to prove that the current implementation maintains CWC. The model setup resembles a single forecast member of DWD's Ensemble Prediction System (EPS). A global domain is used with a horizontal resolution of 40 km and 90 vertical levels reaching as high as $z_{top} = 75$ km. The

model is started from an uninitialized analysis for IAU (Prill et al., 2020). In addition to the 5 prognostic water tracers (see Section 2.1), one passive tracer q_p has been added and initialized with $q_p(x, y, z, t = 0) = 1$ throughout the domain. Please note that we might have chosen any other constant value just as well. This tracer is free from physical sources and sinks, however it takes into account IAU mass tendencies. Hence, the RHS of the tracer mass continuity (1) for q_p is nonzero during the IAU-phase and reads

$$Q_{\rho q_g} = q_g \left.\frac{\partial \rho}{\partial t}\right|_{IAU} .$$

If CWC is maintained, q_p must not deviate from 1 at any point or at any time.

Two model runs with $N_{sub} = 5$ have been conducted: The first run uses the fractional step method as described in Figure 2. The second run uses a variant where the conversion step $\overline{\rho q}^*_{i,k} \longrightarrow \overline{q}^*$ at even time steps is slightly modified. The intermediate density $\overline{\rho}^{**}_{i,k}$ in the denominator of equation (17) is replaced by the density at the new time step $\rho^{n+1}_{i,k}$. This resembles a more naive implementation of the fractional step method without a proper re-integration of the air mass continuity equation. For simplicity, the conversion step for odd timesteps (20) is left unchanged.

Figure 3 shows the domain averaged variance σ^2 of the passive tracer q_p as a function of forecast time. By design σ^2 is zero initially, but it will deviate from zero as soon as q_p starts to deviate from $q_p = 1$ at any grid point. Red and blue curves show results for the unmodified (using $\overline{\rho}^{**}_{i,k}$) and modified (using $\rho^{n+1}_{i,k}$) fractional step method, respectively. Note that the model was initialized at $t = -90$ min in order to obtain a symmetric IAU window.

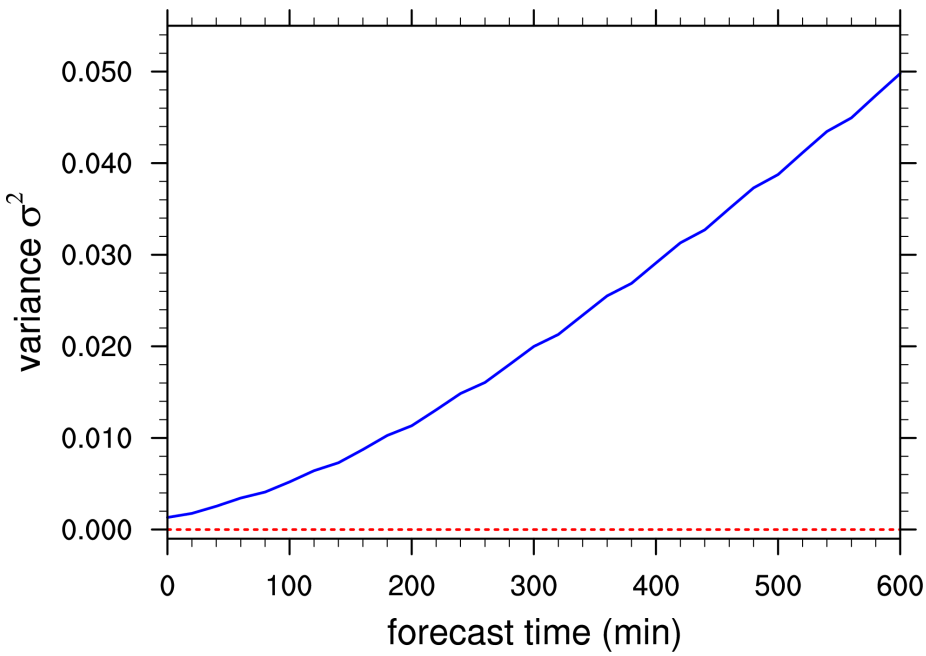

Figure 3: Domain averaged variance σ^2 of the passive tracer q_p as a function of time. Results for the implemented fractional step method (Figure 2) are shown in red. Results for a variant of this method are shown in blue. The latter does not maintain CWC.

In case of the unmodified fractional step method (red curve) σ^2 is zero throughout the entire forecast time. This proofs that tracer transport in ICON maintains CWC exactly. The blue curve, however, starts to deviate from 0 right from the start and keeps increasing throughout the simulation. A snapshot of q_p at some mid-tropospheric model level at $t = 6\,\mathrm{h}$ is shown in Figure 4. Starting from an initially homogeneous field, a pattern of maxima and minima has emerged that keeps increasing over time. This shows that CWC can easily be broken in a fractional step approach, even if identical mass fluxes are enforced for the air mass and tracer mass continuity equation (see Section 3.3). A re-integration of the air mass continuity equation is required in addition.

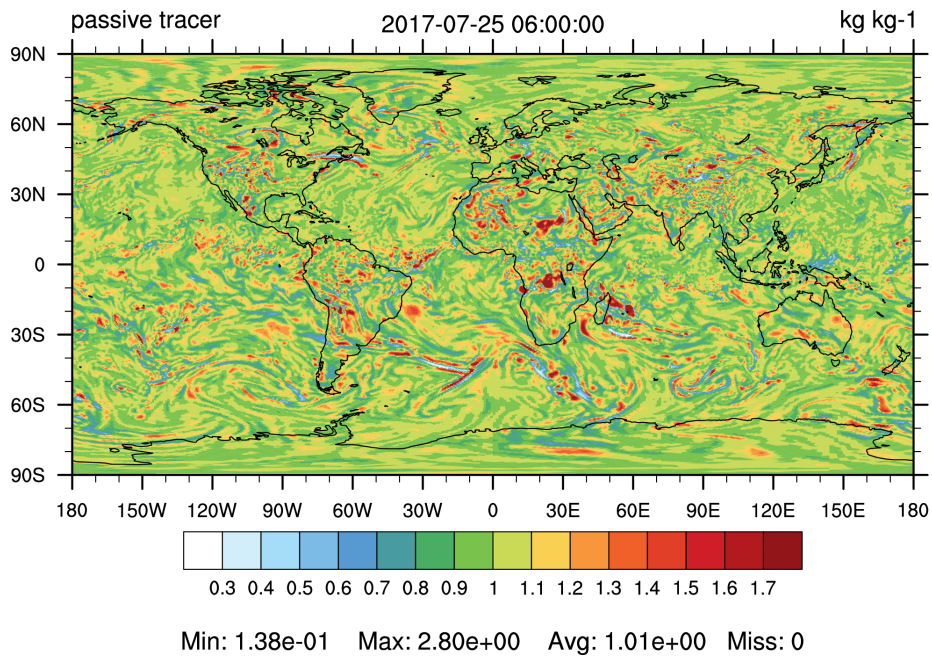

Figure 4: Snapshot of a passive tracer q_p at model level 70 after 6 h, with initial condition $q_p(x, y, z, t = 0) = 1$. A modified variant of the fractional step method has been used for which $\overline{\rho}^{**}_{i,k}$ in equation (17) was replaced by $\rho^{n+1}_{i,k}$.

5. Summary

This article describes the discretization principles and solution procedure of ICON's tracer transport module. Mathematically speaking the transport module solves a first-order linear hyperbolic partial differential equation with source terms, namely the continuity equation for tracer mass. Physically speaking it handles the 3D transport of gaseous, liquid or solid atmospheric constituents due to air motion.

Starting from the continuous form of the tracer mass continuity equation, a cell-integrated form for a prismatic control volume has been derived, which represents the basis of our finite volume discretization. A fractional step (operator splitting) approach has been taken in order to split the 3D cell integrated form into 3 simpler sub-problems, which individually involve the horizontal

and vertical transport operator $\mathcal{H}$ and $\mathcal{V}$, respectively, as well as the source term Q. We have taken special care to maintain consistency between the discretized continuity equations for tracer mass and air mass. To this end we have firstly enforced that the same mass flux is 'seen' by both continuity equations. This has been achieved by reusing the mass fluxes which result from the solution of the continuity equation for air mass. Secondly, we have followed the method of Easter (1993) and incorporated the integration of the air mass continuity equation into the fractional step approach for tracer mass. By this, intermediate changes in tracer density ρq which are solely a result of mass convergence or divergence in the direction of the splitting are removed. This turned out to be essential in order to maintain CWC when fractional steps are used.

To demonstrate the validity of the approach in terms of CWC, a passive tracer q_p with $q_p(x, y, z, t_0) = 1$ has been added to a forecast run with full NWP physics. The tracer field did not deviate from 1 at any point in space or time, which showed that the presented finite volume fractional step approach maintains CWC. Of course this also requires that the applied horizontal and vertical advection operators $\mathcal{H}$, $\mathcal{V}$ are constancy preserving – a point which has not been discussed in this work. Details of the horizontal and vertical advection operators $\mathcal{H}$, $\mathcal{V}$ will be presented in forthcoming reports.

Acknowledgements

The author thanks Günther Zängl (DWD) for many fruitful discussions and useful hints during the development phase. Additional thanks go to Florian Prill (DWD) and Günther Zängl for their careful review of this manuscript and valuable comments.

References

Colella, P., and P. R. Woodward, 1984: The piecewise parabolic method (PPM) for gas-dynamical simulations. *J. Comput. Phys.*, **54**, 174–201.

Donahue, A. S., and P. M. Caldwell, 2018: Impact of physics parameterization ordering in a global atmosphere model. *Journal of Advances in Modeling Earth Systems*, **10 (2)**, 481–499.

Durran, D. R., 1999: *Numerical Methods for Wave Equations in Geophysical Fluid Dynamics*. Springer-Verlag.

Easter, R. C., 1993: Two modified versions of Bott's positive-definite numerical advection scheme. *Mon. Wea. Rev.*, **121**, 297–304.

Gross, E. S., L. Bonaventura, and G. Rosatti, 2002: Consistency with continuity in conservative advection schemes for free-surface models. *International Journal for Numerical Methods in Fluids*, **38 (4)**, 307–327.

Harris, L. M., and P. H. Lauritzen, 2010: A flux-form version of the conservative semi-lagrangian multi-tracer transport scheme (CSLAM) on the cubed sphere grid. *J. Comput. Phys.*, **230**, 1215–1237.

Lauritzen, P. H., C. Erath, and R. Mittal, 2011a: On simplifying 'incremental remap'-based transport schemes. *J. Comput. Phys.*, **230**, 7957–7963.

Lauritzen, P. H., C. Jablonowski, M. A. Taylor, and R. D. Nair, 2011b: *Numerical Techniques for Global Atmospheric Models*. 1st ed., Springer, 556 pp.

Lauritzen, P. H., R. D. Nair, and P. A. Ullrich, 2010: A conservative semi-Lagrangian multi-tracer transport scheme (CSLAM) on the cubed-sphere grid. *J. Comput. Phys.*, **229**, 1401–1424.

LeVeque, R. J., 2002: *Finite-Volume Methods for Hyperbolic Problems*. 1st ed., Cambridge University Press, 558 pp., doi:10.1017/CBO9780511791253.

Lin, S.-J., and R. B. Rood, 1996: Multidimensional flux-form semi-lagrangian transport schemes. *Mon. Wea. Rev.*, **124 (9)**, 2046–2070.

Miura, H., 2007: An upwind-biased conservative advection scheme for spherical hexagonal-pentagonal grids. *Mon. Wea. Rev.*, **135**, 4038–4044.

Prill, F., D. Reinert, D. Rieger, and G. Zängl, 2020: *ICON Tutorial: Working with the ICON model*. Deutscher Wetterdienst (DWD), URL https://www.dwd.de/EN/ourservices/nwv_icon_tutorial/pdf_volume/icon_tutorial2020_en.pdf?__blob=publicationFile&v=3.

Reinert, D., F. Prill, M. Baldauf, and Coauthors, 2020: *DWD Database Reference for the Global and Regional ICON and ICON-EPS Forecasting System*. Deutscher Wetterdienst (DWD), URL https://www.dwd.de/SharedDocs/downloads/DE/modelldokumentationen/nwv/icon/icon_dbbeschr_aktuell.pdf?view=nasPublication&nn=16102.

Skamarock, W. C., and M. Menchaca, 2010: Conservative transport schemes for spherical geodesic grids: High-order reconstructions for forward-in-time schemes. *Mon. Wea. Rev.*, **138**, 4497–4508.

Strang, G., 1968: On the construction and comparison of difference schemes. *SIAM J. Numer. Anal.*, **5 (3)**, 506–517.

Wan, H., and Coauthors, 2013: The ICON-1.2 hydrostatic atmospheric dynamical core on triangular grids Part 1: Formulation and performance of the baseline version. *Geosci. Model Dev.*, **6 (3)**, 735–763.

Zängl, G., D. Reinert, P. Ripodas, and M. Baldauf, 2015: The ICON (ICOsahedral Non-hydrostatic) modelling framework of DWD and MPI-M: Description of the non-hydrostatic dynamical core. *Q. J. R. Meteorol. Soc.*, **141**, 563–579.

Zerroukat, M., N. Wood, and A. Staniforth, 2006: The Parabolic Spline (PSM) Method for conservative transport problems. *Int. J. Numer. Meth. Fluids*, **51**, 1297–1318.